

Kushagra Pathak

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PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer skilled in Machine Learning, Deep Learning, and Generative AI with experience building AI-powered and data-driven applications. Passionate about solving complex problems and building scalable AI solutions with strong analytical and programming skills.

EDUCATION

Babu Banarasi Das Institute of Technology and Management

Bachelor of Technology in Computer Science and Engineering (Data Science)

Lucknow, Uttar Pradesh

Nov 2022 – 2026

TECHNICAL SKILLS

Language & Database: Python, SQL, PostgreSQL, MongoDB

Machine Learning & Deep Learning: Supervised Learning, Regression, Classification, Unsupervised Learning, Clustering, Dimensionality Reduction, Exploratory Data Analysis, Data Preprocessing, Feature Engineering, Model Training, Hyperparameter Tuning, Model Evaluation, Neural Networks, CNNs, RNNs, LSTMs, Transformers

Generative AI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Livekit, Pipecat, SST, TTS, Vector Embedding, Vector Database

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, React.js, FastAPI, Flask, NiceGUI, RESTful API Development

Tools & Version Control: Docker, Git, GitHub

PROFESSIONAL EXPERIENCE

Data Science and AI Trainee

Ducat

Jul 2025 – Aug 2025

Lucknow, India

- Developed a supervised machine learning model using lifestyle and consumption datasets to predict monthly carbon emissions, performing EDA, preprocessing, missing value handling, feature engineering, and One-Hot Encoding.
- Trained and optimized a Random Forest Regressor using Scikit-learn for accurate environmental impact prediction, achieving a Mean Absolute Error (MAE) of 28.38 and an R^2 Score of 99%.
- Deployed the model as a real-time Flask web application for interactive carbon footprint estimation and end-user prediction.

PROJECTS

AI Interview Assistant

Python, FastAPI, Pipecat, LiveKit, SST, LLM, TTS, LangGraph, ChromaDB, PostgreSQL, Docker

- Developed a real-time conversational AI interview assistant capable of conducting voice-based technical and behavioral interviews using Pipecat, LiveKit, Speech-to-Text (Deepgram), Large Language Models (Gemini), and Text-to-Speech for seamless human-like interactions.
- Implemented Retrieval-Augmented Generation (RAG) with ChromaDB to retrieve role-specific interview questions, company knowledge, and evaluation rubrics, enabling contextual conversations, dynamic follow-up questioning, candidate response evaluation, and AI-driven feedback generation.
- Engineered a scalable FastAPI backend with PostgreSQL for interview session management, candidate data, conversation history, scoring, and automated interview report generation, while deploying the application using Docker for production-ready scalability.

AI Product Review Intelligence System

Python, REST APIs, FastAPI, NiceGUI, PostgreSQL, Transformers, LangChain, Selenium, BeautifulSoup, ChromaDB, Docker

- Developed an AI-powered platform to analyze product reviews from e-commerce, YouTube, comparison, and carrier platforms, and built automated web scraping pipelines using Selenium and BeautifulSoup to collect reviews, ratings, comparisons, and creator review data.
- Implemented NLP and Generative AI features including sentiment analysis, AI summarization, fake review detection, and RAG-based semantic search using Transformers and LangChain.
- Designed an interactive NiceGUI dashboard for product search, competitor comparison, sentiment analytics, and AI-generated insights, while integrating MongoDB and ChromaDB for scalable data storage, semantic retrieval, vector embeddings, and AI chatbot support.

Carbon Footprint Estimator

Python, Scikit-learn, Pandas, NumPy, Flask, Pickle, AST

- Developed a supervised machine learning model to predict monthly carbon emissions (kg/month) based on lifestyle habits, achieving a 99% R^2 score using a RandomForestRegressor.
- Performed extensive Exploratory Data Analysis (EDA), implemented a random-fill imputation strategy for categorical null values, and applied feature engineering using AST and One-Hot Encoding for complex categorical inputs.
- Trained and evaluated the model to achieve a Mean Absolute Error (MAE) of 28.38, and developed a Flask web interface to serve the serialized Pickle model for real-time carbon footprint estimation.

CERTIFICATIONS

- Data Science and AI Certification** — Ducat ([View Credential](#))
- Python Certification** — GUVI | HCL ([View Credential](#))
- Machine Learning Certification** — GUVI | HCL ([View Credential](#))
- SQL Certification** — GUVI | HCL ([View Credential](#))